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Research paper

Auto-encoding clinical language for zero-shot medical report generation

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ABSTRACT

Medical report generation, which aims to generate clinical descriptions for medical images, has attracted extensive research interest recently. However, conventional methods require a large-scale labeled dataset for training. In medical scenarios, sufficient image-report pairs are usually unavailable, especially for rare and novel diseases. To address this challenge, we propose an effective and intuitive zero-shot learning model, the auto-encoding clinical language model for zero-shot medical report generation (ZeroMRG), which aims to reduce the dependency on labeled data. Unlike previous approaches that require separate models for different tasks, our model is capable of handling medical report generation tasks across multiple languages and modalities. The core strength of our model lies in its robust zero-shot learning capability. At the training phase, we introduce the auto-encoding method, which reconstructs the input reports, to learn to generate reports based on the textual embeddings from diverse languages and modalities. During the testing phase, we employ prompt learning and Contrastive Language–Image Pre-training (CLIP) techniques to create visual embeddings that align seamlessly with textual embeddings. The aligned embeddings are then fed into the decoder to directly generate reports for medical images, without the need for training on labeled image-report pairs. We conduct experiments on three datasets, covering Chinese and English languages and spanning two different medical imaging modalities. The results show that our model can achieve over 90% of the performance of state-of-the-art supervised models. This suggests the potential of our model for rare diseases in low-resource settings.

1. Introduction

Medical images, e.g., chest X-ray and pathology images, along with their corresponding reports that provide detailed descriptions of both normal and abnormal regions, are widely used for diagnosis and treatment (Goergen et al., 2013; Pool and Goergen, 2010). Physicians normally write reports according to the syndromes of patients that are reflected in medical images so as to form professional records. While, writing medical reports is a time-consuming job, and error-prone for inexperienced physicians, which motivated a series of studies (Liu et al., 2022a, 2021a,b,c; Monshi et al., 2020; Qin and Song, 2022; Selivanov et al., 2023; Tanida et al., 2023; Zhang et al., 2020) on automatic report generation. Automatically generating medical reports can assist physicians in clinical decision-making. Researchers have achieved great success on this topic, which has emerged as a prominent attractive research direction in both artificial intelligence and clinical medicine.

The generation of medical reports, a crucial component of diagnostic processes, has seen a significant influx of technological advancements in recent years. Many existing models adopt the standard encoder–decoder approach, which incorporates a visual encoder to interpret the image inputs and a text decoder to generate the desired reports. However, the dependency of these models on extensive annotated datasets for training poses a significant challenge in medical contexts, where the availability of labeled data is usually limited. Medical scenarios present unique challenges due to the fact that data must be manually labeled by experienced clinicians, which is a time-consuming and costly process. Besides, these scenarios often involve sensitive and private health information, necessitating strict adherence to data privacy and security measures. Furthermore, the datasets commonly utilized for medical report generation models, such as COVID-19 CTR with 0.7k samples (Li et al., 2023b), COVID-19 CT with 0.3K samples (Liu et al., 2021d), and IU-Xray with 7k samples (Demner-Fushman et al., 2016) are relatively small when compared to the conventional datasets like Conceptual Captions with its 3.3M samples used for image captioning tasks. In the case of rare and novel diseases, including emerging pandemics and variants, obtaining sufficient labeled data for training purposes can be challenging, especially during the early stages. Nevertheless, it is precisely during these critical periods that AI assistance tools are arguably most urgently required (Liu et al., 2023). Additionally, the acquisition of non-English medical image

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datasets is quite difficult. Due to the diversity and specialization of the languages involved, processing these datasets poses even greater challenges. Moreover, the traditional approaches often relied on separate, specialized models for distinct medical tasks, leading to high computational costs and a lack of comprehensive solutions.

To this end, we propose an novel approach, the auto-encoding clinical language model for zero-shot medical report generation (ZeroMRG), which has practical value for English and non-English medical report generation where data-text pairs are limited in availability. The core of ZeroMRG lies in its powerful zero-shot learning capabilities, enabling the model to generate high-quality medical reports without requiring any labeled data for training. During the training phase, we seamlessly incorporate the concept of auto-encoding, with the objective of establishing cross-modal (e.g., chest X-rays and CT scans) and cross-lingual (e.g., English and Chinese) mappings. This integration is intended to bridge the gap between these diverse data sources, unifying them within a single feature space. The ultimate goal of this design is to dismantle barriers that exist across different modalities and languages, thereby facilitating the seamless sharing and fusion of information. During the inference phase, our model is tasked with generating medical reports solely based on the input medical images. We utilize prompt learning in conjunction with CLIP techniques to generate semantically rich visual embeddings. The utilization of prompt learning serves as a vital bridge that significantly narrows the gap between the textual and visual modalities. These optimized visual embeddings incorporate not only the visual information inherent to the images, but also the relevant textual information, offering a richer and more nuanced understanding of the images. Subsequently, these enriched visual embeddings are fed into the decoder, which directly generates the medical reports.

Unlike existing approaches, which often rely on specialized models tailored to specific tasks or patterns, ZeroMRG is designed as a general solution. It is capable of handling multimodal and multilingual medical report generation, demonstrating its robustness and adaptability to various medical imaging scenarios. To validate the effectiveness of ZeroMRG, we conduct extensive experiments on three datasets, COVID-19 CTR (COV-CTR), COVID-19 CT (COV-CT) and IU-Xray. The IU-Xray dataset encompasses chest radiographs along with their English reports. The COV-CTR and COV-CT datasets consist of computed tomography (CT) scans and their corresponding English and Chinese reports, respectively. By utilizing these three datasets to evaluate our model, we ensure a comprehensive assessment of ZeroMRG's cross-lingual and cross-modal capabilities. The experimental results demonstrate that ZeroMRG is able to generate high-quality and credible medical reports without relying on any labeled data for training. Furthermore, ZeroMRG can achieve exceptional zero-shot performance and is comparable to state-of-the-art fully supervised methods, highlighting its superiority and potential for practical applications in the medical domain.

To sum up, our contributions are as follows:

- We propose an efficient model, the ZeroMRG, for zero-shot medical report generation. The model can generate high-quality medical reports across multiple modalities and languages, without any downstream labeled image-text pairs for training.
- We evaluate the effectiveness of ZeroMRG using the COVID-19 pandemic "in replay". We provide a comprehensive evaluation of our proposed ZeroMRG on two COVID-19 datasets and IU-Xray dataset in both Chinese and English languages.
- The results demonstrate that our method not only achieves superior zero-shot performance, but also compares favorably with previous fully-supervised methods. Our model is particularly advantageous for rare and novel diseases, where the labeled data for training are usually very limited or even unavailable. This technological innovation has the potential to facilitate the widespread adoption and application of medical artificial intelligence technology.

Our proposed ZeroMRG represents a significant advancement in the field of medical report generation. By leveraging zero-shot learning, multi-modal auto-encoding, CLIP techniques and prompt learning, it overcomes the limitations of traditional methods that rely heavily on labeled data. In the subsequent sections, we will present our methodology, experimental setup, results, and comprehensive analysis.

2. Related work

The most popular related task to medical report generation is image captioning (Cornia et al., 2020a; Pan et al., 2020; Tewel et al., 2022; Wang et al., 2019; Xu et al., 2015; Yang et al., 2019), which aims to generate captions to describe the given images. In contrast to the traditional image captioning, medical report generation is a challenging application that demands significantly longer outputs and incorporates unique features, thus requiring tailored solutions that address its unique challenges.

2.1. Medical report generation

Nowadays, clinical medicine has attracted increasing attention, and several works have explored different approaches and techniques for medical report generation. Most existing medical report generation works (Shin et al., 2016; Wang et al., 2018; Zhang et al., 2020) attempted to adopt CNN-LSTM based models to automatically generate fluent reports. However, due to the limited availability of medical data, these models are prone to generating plausible yet generic reports that fail to highlight prominent abnormal narratives. With the introduction of the attention mechanism (Xu et al., 2015), several studies in the field of medical image analysis started incorporating attention to medical report generation (Jing et al., 2017; Yuan et al., 2019; Zhang et al., 2017). Some studies attempted to use transformer-based models as decoders in the medical report generation domain (Alfarghaly et al., 2021; Chen et al., 2020). Some works utilized useful visual and textual features (Li et al., 2018; Tanida et al., 2023) into medical report generation. To better align different information and guide the generation process, recent studies are focusing on improving cross-modal alignment through reinforcement learning (Liu et al., 2019; Qin and Song, 2022), as well as memory networks (Chen et al., 2022b, 2020).

Recently, with the remarkable generative capabilities of large language models (LLMS) (Javaheripi et al., 2023; Touvron et al., 2023; Waisberg et al., 2023), some studies have employed LLMs to multimodal scenarios (Chen et al., 2022a; Zhu et al., 2023), including medical domain (Li et al., 2023a; Liu et al., 2024a; Thawkar et al., 2023; Tu et al., 2023). However, the direct application of LLMs to the medical domain often leads to inaccuracies or limited effectiveness due to the existence of a domain gap between general corpora and medical corpora. To counter this problem, Lee et al. (2020) pretrained BERT (Devlin et al., 2018) on medical corpora.

2.2. Zero-shot learning

The realms of vision and text are inherently disparate, with each domain possessing distinct modalities that exhibit unique features and nuances. This profound divergence, along with the specificity inherent in these various forms, poses significant challenges for zero-shot learning. As a result, the works dedicated to zero-shot image captioning (Li et al., 2023c; Liu et al., 2022b; Nukrai et al., 2022; Tewel et al., 2022; Yang et al., 2024; Zeng et al., 2022) remain relatively scarce. For example, Gu et al. (2018) proposed a language pivoting method that harnesses the characteristics of an image captioner trained on a pivot language (Chinese) and aligns it to a target language (English) through a pivot-target (Chinese-English) parallel corpus, without requiring training on the downstream vision-text pairs. Zeng et al. (2022) adopted LLMs like GPT (Brown et al., 2020) for zero-shot image captioning tasks. Although this technique demonstrates notable efficacy, it lacks adaptability in multilingual environments. Yang et al. (2024) proposed the ZeroNLG, a zero-shot learning framework that enables cross-modal and cross-lingual natural language generation without requiring labeled downstream data. However, it is noteworthy that this model, despite its remarkable effectiveness, is not specifically tailored for the medical reports generation.

As a branch of the image captioning task, the application of zero-shot learning to medical report generation is relatively uncommon, especially when it comes to the realm of multi-language and multi-modal medical report generation. This area remains a relatively under-researched frontier and poses even greater challenges for zero-shot learning. Hirsch et al. proposed MedRAT (Hirsch et al., 2024), a framework that utilizes unpaired image and report data to generate medical reports by combining auto-encoding with multi-modal alignment, facilitated by auxiliary tasks of contrastive learning and classification. Liu et al. (2024b) proposed MCVGen, a zero-shot medical report framework using multimodal contextual vectors for efficient knowledge transfer from limited examples. Huang et al. (2024) introduced Maco, a contrastive X-ray foundation model that achieves both fine-grained understanding and zero-shot learning via correlation-weighted masking. There are other zero-shot works for medical report generation like (Bhardwaj et al., 2025; Hu et al., 2024). However, these methods have a notable limitation in their adaptability to both multilingual and multimodal environments.

To this end, to systematically address the key limitations identified in prior works: the heavy reliance on paired image-report data and the notable limitation in adaptability to both multilingual and multimodal environments, we propose the ZeroMRG model. ZeroMRG is designed to tackle a diverse range of medical report generation issues across multimodal and multilingual contexts within a unified framework, and it possesses the capability to simultaneously generate multimodal and multilingual medical reports. Furthermore, ZeroMRG eliminates the need for paired training data, while still achieving comparable performance to fully supervised methods. Consequently, ZeroMRG has the potential for widespread adoption, especially in diverse and resource-constrained settings.

3. Method

In this section, we introduce our proposed ZeroMRG. An overview of our proposed model is shown in Fig. 1, where the details are illustrated in the following subsections.

3.1. Formulation

The task of medical report generation aims to understand the input medical image I and subsequently generate a topic related report $S = \{s_1, s_2, \ldots, s_T\}$ which including T words. The report is composed of multiple sentences, each aiming to provide a descriptive account of the patient's condition reflected in the medical image.

As shown in Fig. 1, we perform unsupervised auto-encoding to learn how to generate reports by reconstructing the input reports. Subsequently, we utilize the trained model to infer from medical images. In implementations, we choose English reports and Chinese reports to evaluate our approach. We denote the English report as S_{en}, and the Chinese report as S_{cn}. The CT scan image is denoted as I_{ct}, and the X-ray image is denoted as I_{ctr}. ZeroNLG includes three modules, i.e., a multilingual encoder E_m, a multilingual decoder D_m, and a vision encoder E_v. Our ZeroNLG is defined as follows:

$$\begin{aligned} \text{Auto Encoding}: E_m(S_{en}) &\xrightarrow[\mathrm{D}_m(.)]{\text{reconstruction}} S_{en}, \\ \text{Auto Encoding}: E_m(S_{cn}) &\xrightarrow[\mathrm{D}_m(.)]{\text{reconstruction}} S_{cn}, \\ \text{Zero-shot Vision-to-Text}: E_v(I) &\xrightarrow[\mathrm{D}_m(.)]{\text{Generation}} S_j, \end{aligned} \tag{1}$$

where $i, j = 1, 2, 3, 4, \ldots$; ZeroMRG can capture the inherent relationships between medical images and their corresponding textual descriptions in multi-modal and cross-lingual scenarios. Given the ground truth report, $S^*_{en} = \{s^*_{en_1}, s^*_{en_2}, \ldots, s^*_{en_T}\}$ and $S^*_{cn} = \{s^*_{cn_1}, s^*_{cn_2}, \ldots, s^*_{cn_T}\}$ for the input report, we can train our model by minimizing a widely-used cross-entropy (XE) loss:

$$L_{\mathrm{XE}}(\theta) = -\sum_{t=1}^{T} \log\left(p_\theta\left(s^*_{en_t} \mid s^*_{en_{1:t-1}}\right) * \left(s^*_{cn_t} \mid s^*_{cn_{1:t-1}}\right)\right). \tag{2}$$

The parameter θ represents the model's parameters, and p_θ denotes the probability that how likely the corresponding word in the vocabulary is the current output word. This loss function allows the model to learn to generate accurate reports by minimizing the discrepancy between the predicted and ground-truth reports.

Our model can be partitioned into two major components: text-only training and zero-shot inference. These components are detailed below.

3.2. Text-only training

During the training phase, we only have access to a set of medical reports. To process these reports, we utilize auto-encoding technology to reconstruct the input text from the textual embeddings generated by Multilingual-CLIP (Carlsson et al., 2022).

In the encoder phase (i.e., E_v), we use Multilingual-CLIP, a widely adopted multilingual version of Contrastive Language–Image Pre-training (CLIP) (Radford et al., 2021) capable of managing various languages, to extract textual features $\psi(S)$ from the medical reports, regardless of their language. To eliminate the scale discrepancies, we normalize the textual features $\psi(S)$, with the normalization defined as follows:

$$\psi(S) = \frac{\psi(S)}{\|\psi(S)\|_2}, \tag{3}$$

where $\|\psi(S)\|_2$ denotes the ℓ_2-norm of the textual embedding. This process captures semantic and contextual information within the reports, allowing us to compare and process reports in different languages within the same feature space. In the decoder phase, we decode the textual features back into the medical reports. Our objective is to reconstruct the output reports as closely as possible to the original reports. This decoding process ensures that the decoded text accurately captures and reflects the semantic content and essential context of the original medical report, thereby preserving its critical information. The encoder and decoder processes can bridge the gap between different data sources, unify them into a single feature space, break down barriers between different modalities and languages, and promote seamless sharing and integration of information. The process can be defined as follows:

$$(S_{\mathrm{en}}, S_{\mathrm{cn}}) \to D_m\left(E_v\left(S_{\mathrm{en}}, S_{\mathrm{cn}}\right)\right) \to \left(S_{\mathrm{en}}, S_{\mathrm{cn}}\right). \tag{4}$$

3.3. Zero-shot inference

In the ZeroMRG, zero-shot inference serves as the key to realize unsupervised medical report generation. The innovative mechanism leverages the trained D_m to generate reports directly for the never-before-seen medical images.

Given an image I, we first get the image embedding $\varphi(I)$ by Multilingual-CLIP and prompts. The Multilingual-CLIP model and prompts are "frozen" during the this process. The formulation is below:

$$\varphi(I) = \frac{f_{\mathbf{M}}(I) + f_{\mathbf{p}}(I)}{\|f_{\mathbf{M}}(I) + f_{\mathbf{p}}(I)\|_2}, \tag{5}$$

$$f_{\mathbf{p}}(I) = \mathrm{softmax}\left(\frac{f_{\mathbf{M}}(I)\left(f_{\mathbf{p}}\right)^\top}{\sqrt{d}}\right) f_{\mathbf{p}}, \tag{6}$$

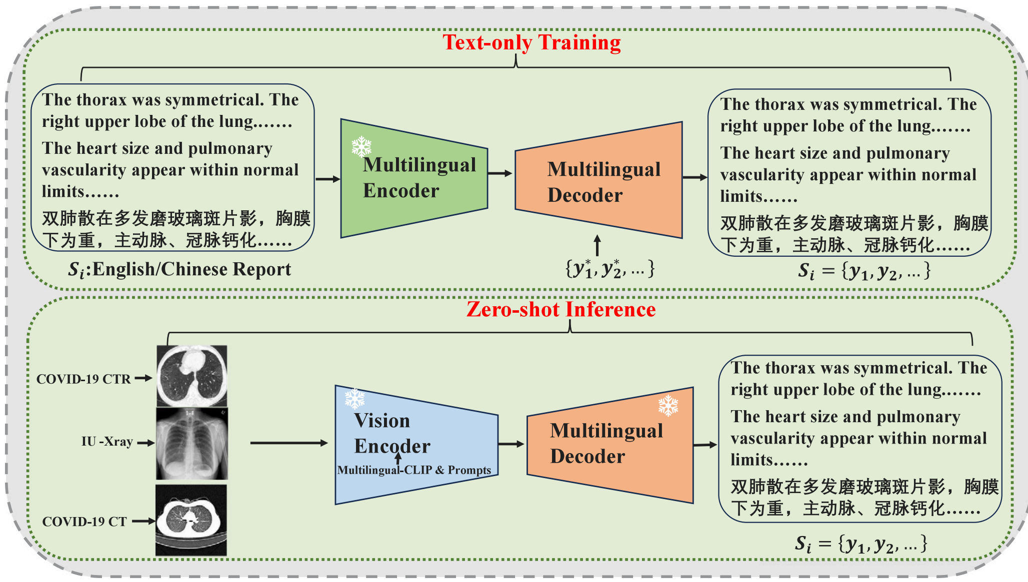

Fig. 1. The illustration of our proposed ZeroMRG.

where $f_{\mathbf{M}}(I)$ is the visual embeddings extracted by Multilingual-CLIP, $f_{\mathbf{p}}(I)$ is the prompt-based visual embeddings, $f_{\mathbf{p}}$ is the prompt embeddings; $f_{\mathbf{M}}(I) \in \mathbb{R}^{N_M \times d_M}, f_{\mathbf{p}} \in \mathbb{R}^{N_p \times d_p}$ $(d_M = d_p)$; $\|f_{\mathbf{M}}(I) + f_{\mathbf{p}}(I)\|_2$ denotes the ℓ_2-norm of $f_{\mathbf{M}}(I) + f_{\mathbf{p}}(I)$. By incorporating the power of pre-trained Multilingual-CLIP, ZeroMRG is able to extract meaningful visual embeddings from medical images. The embeddings can capture the essential features and information present in the images, serving as the foundation for generating medical reports. However, the domain gap between image and text modalities is a significant challenge in tasks like medical report generation, especially when using pre-trained models like Multilingual-CLIP that were trained on general-purpose image-text pairs. To mitigate this domain gap, we adopt prompt learning (Gu et al., 2021; Zhou et al., 2022), which has been widely employed in current state-of-the-art large language models. Many studies show that the constructed prompts can preserve a knowledge base or a working memory for inference (Kumar et al., 2016; Vinyals et al., 2016). By designing prompts that are tailored to the medical domain and relevant to report generation, ZeroMRG is guided to focus on specific aspects of the images and generate more precise and targeted reports. We randomly select n medical reports as prompts, and incorporate these prompts into the visual embeddings generated by Multilingual-CLIP, thereby creating a new set of enriched visual embeddings. Then, we apply the previously trained decoder D_m to these embeddings $\varphi(I)$ to generate the corresponding descriptions.

Our proposed approach provides a new solution for the medical report generation by combining the Multilingual-CLIP with prompt learning. By eliminating the need for labeled data, ZeroMRG significantly expands the applicability of medical report generation.

3.4. Fully-supervised training

To verify the generalization ability of the model and further enhance the model's performance and accuracy, we conduct fully supervised training on the basis of auto-encoding. Unlike zero-shot inference, where the module is directly applied to new tasks without additional training, we not only train the encoder E_v to extract features from input data, but also train the decoder D_m to generate reconstructions or predictions based on the encoder's output. By utilizing labeled dataset for training, fully-supervised training can improve the model's accuracy and performance compared to zero-shot inference. However, this method also relies on a large amount of high-quality labeled data, and thus may be limited in certain application scenarios.

We train our model by minimizing a widely-used cross-entropy (XE) loss function. The loss function is designed to facilitate the model in improving its report generation accuracy by minimizing the discrepancies between the predicted and ground-truth reports. Specifically, given the ground truth report for the image, the loss function is defined as follows:

$$L_{\mathrm{XE}}(\gamma) = -\sum_{t=1}^{T} \log \left(p_\gamma \left(s^*_{en_t} \mid s^*_{en_{1:t-1}}; \varphi(I) \right) * \left(s^*_{cn_t} \mid s^*_{cn_{1:t-1}}; \varphi(I) \right) \right). \quad (7)$$

4. Experimental results and analysis

In this section, we describe three public datasets along with some commonly used metrics and experimental settings. Then we present the evaluation and analysis of the proposed approach, showcasing its effectiveness in generating reports in the medical scenarios.

4.1. Datasets

We conduct our experiments on three datasets, i.e., COV-CTR (Li et al., 2023b), COV-CT (Liu et al., 2021d) and IU-Xray (Demner-Fushman et al., 2016).

COV-ctr. The COV-CTR dataset consists of 728 images, with 349 images depicting COVID-19 cases and 379 images representing non-COVID cases. These images were sourced from published papers and are paired with corresponding English and Chinese reports. In this study, we utilize only the English reports. We randomly split the dataset into training, validation, and testing sets in a ratio of 8 : 1 : 1, ensuring no overlap among these subsets.

COV-CT. The COV-CT dataset specifically focuses on the collection of chest CT images accompanied by their corresponding Chinese medical reports, designed primarily for the purpose of COVID-19 diagnosis. This collection comprises 368 medical reports and 1104 CT images, derived from 96 patients at The First Affiliated Hospital of Jinan University in Guangzhou, China, and The Fifth Affiliated Hospital of Sun Yat-sen University in Zhuhai, China. Specifically tailored for report generation, each report in this dataset is accompanied by ten CT images, each of which captures distinct features and has been carefully selected to reflect the distinctive medical findings highlighted in the corresponding narratives. We partition the dataset randomly into training, validation, and testing subsets in 8:1:1 ratio.

Table 1
Performance of our approach on English medical report generation (COV-CTR dataset) trained with different ratios of labeled image-report pairs.

Methods	Ratio of data	English report generation (COV-CTR Li et al., 2023b)					
		BLEU-1	BLEU-2	BLEU-3	BLEU-4	CIDEr	ROUGE-L
CoAtt (Jing et al., 2017)	100%	70.9	64.5	60.8	55.2	67.2	74.8
NIC (Vinyals et al., 2015)	100%	69.7	62.1	56.8	51.5	65.9	72.3
AdaAtt (Lu et al., 2017)	100%	67.6	63.3	59.6	51.4	68.2	72.6
Vision-BERT (Devlin et al., 2018)	100%	71.0	65.3	60.6	55.8	68.4	74.7
Vision-GPT (Radford et al., 2018)	100%	70.8	64.5	60.0	54.9	68.0	74.6
R2Gen (Chen et al., 2020)	100%	71.6	63.8	57.6	52.4	152.8	67.7
ASGK (Li et al., 2023b)	100%	71.2	65.9	61.1	57.0	68.4	74.6
	0%	**65.9**	**58.5**	**54.8**	**50.7**	**63.5**	**67.4**
Ours	**10%**	**68.8**	**62.1**	**59.5**	**53.6**	**66.5**	**72.3**
	100%	**75.1**	**68.6**	**64.4**	**61.2**	**72.3**	**77.5**

Table 2
Performance of our approach on Chinese medical report generation trained with different ratios of labeled image-report pairs.

Methods	Ratio of data	Chinese report generation (COV-CT Liu et al., 2021d)					
		BLEU-1	BLEU-2	BLEU-3	BLEU-4	CIDEr	ROUGE-L
CoAtt (Jing et al., 2017)	100%	60.8	53.5	49.4	46.8	25.5	57.4
Vision-BERT (Devlin et al., 2018)	100%	57.9	51.4	47.5	44.9	29.6	57.8
DenseNet+GPT-2 (Liu et al., 2021d)	100%	55.3	47.6	42.3	39.0	23.1	51.4
VGG+GPT-2 (Liu et al., 2021d)	100%	55.8	48.0	42.6	39.2	31.5	51.3
	0%	**58.8**	**51.4**	**46.2**	**41.9**	**42.8**	**50.7**
Ours	**10%**	**62.9**	**54.8**	**50.5**	**47.6**	**46.7**	**55.1**
	100%	**69.4**	**61.2**	**56.6**	**53.5**	**57.3**	**60.9**

IU-xray. IU-Xray (Demner-Fushman et al., 2016) is a widely-used benchmark dataset to evaluate the performance of radiology report generation methods. It contains 7470 chest X-ray images associated with 3955 radiology reports. All the reports enclose the following sections: impression, findings, tags, comparison, and indication. We use the concatenation of impression and findings as the target reports. We apply the conventional random split method, ensuring a 70%-10%-20% distribution for the training, validation, and testing sets.

4.2. Evaluation metrics

To assess the quality of the generated medical reports, we employ the standard and commonly used evaluation metrics, including CIDEr (Vedantam et al., 2015), ROUGE-L (Lin, 2004), and BLEU (Papineni et al., 2002). These metrics essentially evaluate the coherence between the n-gram occurrences in the generated reports and the reference reports, taking into account the significance and the rarity of these n-grams. All these metrics are calculated by the standard evaluation toolkit (Chen et al., 2015). Specifically, BLEU (Papineni et al., 2002) was originally proposed for machine translation evaluation. ROUGE-L (Lin, 2004) was designed for measuring the quality of summaries. CIDEr (Vedantam et al., 2015) was designed to evaluate image captioning systems.

4.3. Experimental settings

During training, we set the mini-batch size to 6, as a balance between computational efficiency and the desired randomness. We use dropout and early stopping to avoid overfitting. We early stop if the validation BLEU score (Papineni et al., 2002) has not improved over the last 50 epochs. Our entire framework is trained in an end-to-end way under cross entropy loss with SGD optimizer (Bottou et al., 2018), which is a stochastic gradient descent method with an learning rate of 1e−3 and a momentum value of 0.99. To ensure that reports are not too long or too short, the report generation process would be halted until a special END token is predicted or a predefined max sentence length is reached. In the auto-encoding phase, we extract text features using the Multilingual-CLIP model named "M-CLIP/XLM-Roberta-Large-Vit-L-14". This model combines cross-lingual capabilities with robust language comprehension abilities, enabling it to capture rich semantic information from multi-lingual texts. We use a transformer-based (Vaswani et al., 2017) mapping network for the decoder architecture. We adopt a random selection of the reports from the training dataset to obtain the prompts. The number of prompts used for COV-CTR, COV-CT and IU-Xray are 500, 290, and 1000 respectively.

4.4. Main results

Tables 1, 2, and 3 show the results of the experiments that are carried out using a 0% training set, 10% training set and a 100% training set on the COV-CTR (Li et al., 2023b), COV-CT (Liu et al., 2021d), and IU-Xray (Demner-Fushman et al., 2016). We report the following metrics: BLEU (Papineni et al., 2002), METEOR (Denkowski and Lavie, 2014), ROUGE-L (Lin, 2004), and CIDEr (Vedantam et al., 2015) which were found to better correlate with human judgments.

Based on the experimental results, we can conclude that our model, without any training data, can achieve over 90% performance to the base models trained with 100% of the paired, supervised data. When using the ZeroMRG model with 10% of the training data, we observe an upward trend in all metrics. Even though the increases are modest, nearly every metric is able to match or even surpass the performance of the fully supervised methods. Additionally, when using the ZeroMRG model with 100% of the training data, we observe an improvement in each evaluation metric. Specifically, the improvement for the COV-CTR dataset is approximately 4%, the improvement for the COV-CT dataset is approximately 9%, and the improvement for the IU-Xray dataset is approximately 3%. This suggests that our approach, which does not rely on direct supervision, has the potential to generalize well to similar tasks and fully supervised tasks. This finding opens up promising new possibilities for AI research and applications, particularly in domains where data is limited or expensive to be acquired.

4.5. Quantitative analysis

We conduct the quantitative analysis to assess the individual contribution of CLIP techniques and prompts to our model's performance, using the COV-CT dataset. The results are presented in Table 4. We

Table 3
Performance of our approach on English medical report generation (IU X-ray dataset) trained with different ratios of labeled image-report pairs.

Methods	Ratio of pairs	English report generation (IU X-ray Demner-Fushman et al., 2016)					
		BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE-L
NIC (Vinyals et al., 2015)	100%	21.6	12.4	8.7	6.6	–	30.6
AdaAtt (Lu et al., 2017)	100%	22.0	12.7	8.9	6.8	–	30.8
Att2in (Rennie et al., 2017)	100%	22.4	12.9	8.9	6.8	–	30.8
Transformer (Vaswani et al., 2017)	100%	39.6	25.4	17.9	13.5	16.4	34.2
$\mathcal{M}^2$ Trans. (Cornia et al., 2020b)	100%	43.7	29.0	20.5	15.2	17.6	35.3
R2Gen (Chen et al., 2020)	100%	47.0	30.4	21.9	16.5	18.7	37.1
R2GenCMN (Chen et al., 2022b)	100%	47.5	30.9	22.2	17.0	19.1	37.5
PPKED (Liu et al., 2021a)	100%	48.3	31.5	22.4	16.8	19.0	37.6
Ours	**0%**	**40.8**	**25.6**	**16.5**	**13.3**	**14.7**	**33.8**
	10%	**43.9**	**29.8**	**19.5**	**15.4**	**17.6**	**35.5**
	100%	**50.5**	**35.3**	**26.2**	**18.6**	**21.4**	**40.8**

Table 4
Ablation study in terms of the utilization of CLIP techniques and prompts for visual embedding generation in zero-shot setting, few-shot learning setting, and fully-supervised setting.

Ratio of pairs	Multilingual-CLIP	Prompts	Chinese report generation (COV-CT Liu et al., 2021d)					
			BLEU-1	BLEU-2	BLEU-3	BLEU-4	CIDEr	ROUGE-L
0%	√	–	40.2	35.9	31.9	27.6	20.1	28.4
0%	–	√	49.4	45.0	40.7	36.6	38.1	44.5
0%	√	√	**58.8**	**51.4**	**46.2**	**41.9**	**42.8**	**50.7**
10%	√	–	48.8	43.5	38.2	34.6	33.0	41.1
10%	–	√	58.4	51.3	47.4	43.4	41.8	52.2
10%	√	√	**62.9**	**54.8**	**50.5**	**47.6**	**46.7**	**55.1**
100%	√	–	53.8	48.0	45.6	42.8	49.6	52.4
100%	–	√	59.5	55.3	53.2	49.5	53.4	56.6
100%	√	√	**69.4**	**61.2**	**56.6**	**53.5**	**57.3**	**60.9**

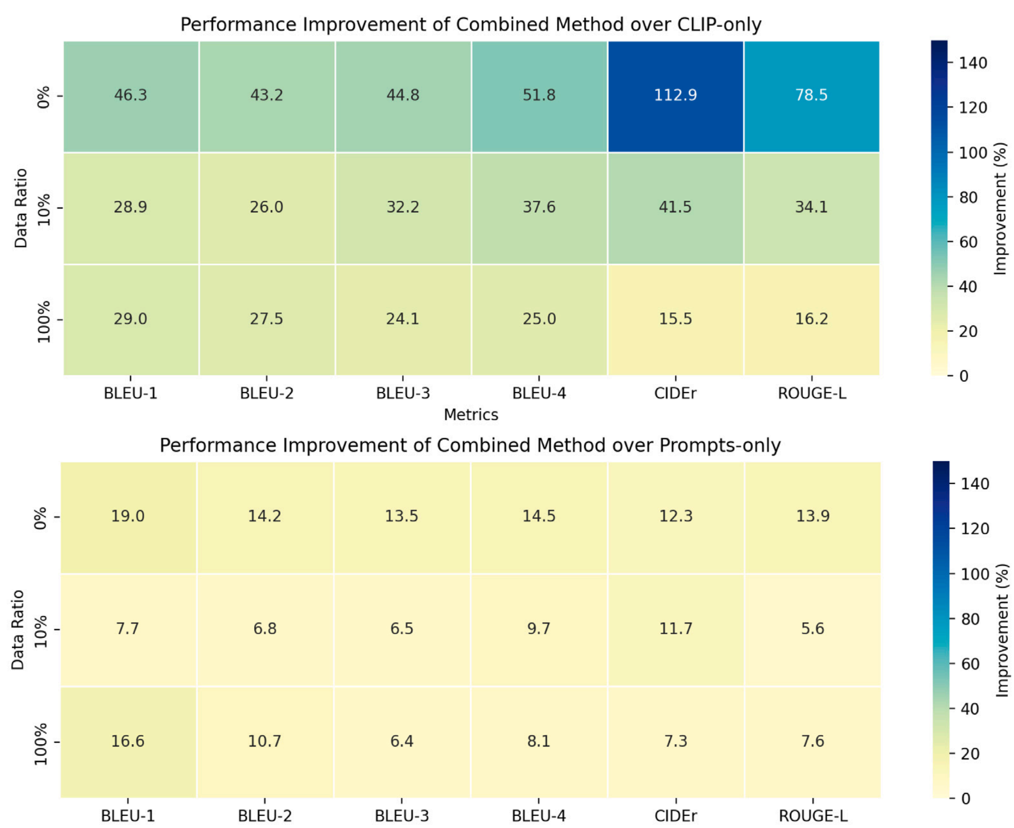

Fig. 2. Performance improvement of combined method over Multilingual-CLIP only and prompts only.

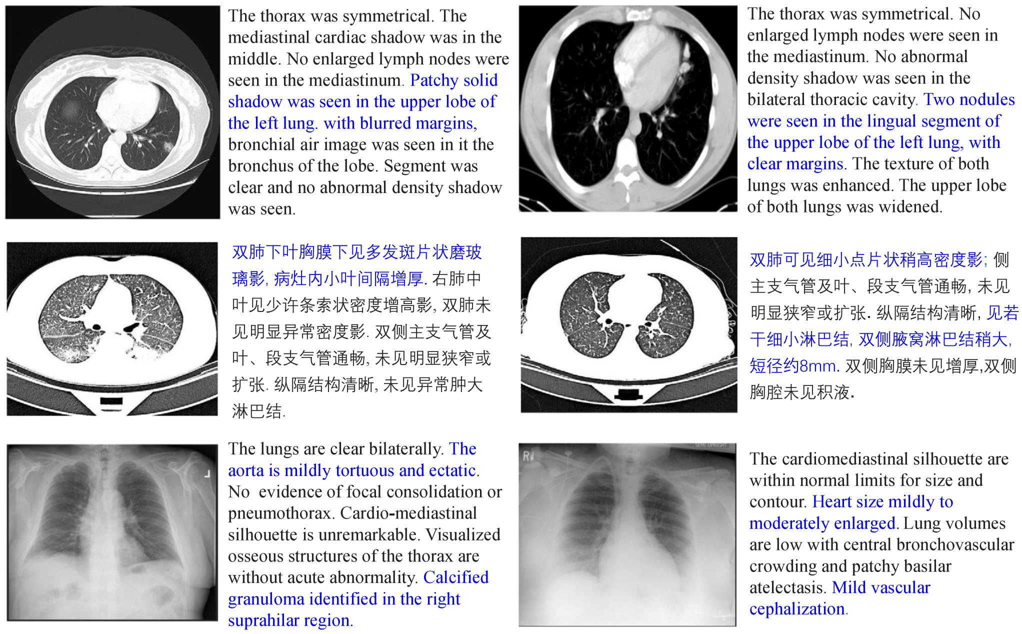

Fig. 3. The examples of Chinese and English medical reports generated by ZeroMRG.

perform those analyses in zero-shot, few-shot, and fully-supervised settings, aiming to gain a deeper understanding of how CLIP techniques and prompts improve model performance under different conditions. Our findings suggest that both CLIP techniques and prompts are capable of enhancing performance, with prompts achieving superior results in comparison. We acknowledge that the direct application of CLIP techniques faces a significant modal gap between images and reports, which cannot effectively optimize on its own. However, prompts enriched with knowledge can significantly aid the model in learning to compose high-quality reports. By leveraging previous expert-written reports and incorporating image information, these prompts can effectively bridge the gap. Ultimately, we discover that directly combining CLIP techniques and prompts results in an overall improvement across all metrics.

We further provide the heatmaps (as shown in Fig. 2) to visualize the relative performance improvement of the combined CLIP and prompts method compared to individual CLIP and prompts for visual embedding generation across different data regimes. Results are shown for six metrics, with color intensity representing the magnitude of improvement. The upper heatmap shows gains over CLIP-only baselines, while the lower heatmap demonstrates improvements against prompts-only settings, allowing direct comparison of how each component contributes to the final performance. We can find that the combined method achieves significantly higher improvements over CLIP-only baselines compared to improvements over prompts-only settings. This is particularly evident in zero-shot settings, where the CIDEr metric improves by 112.9%. This highlights the significant modality gap that exists in Multilingual-CLIP when it is used without textual prompts. Performance deltas decrease as more training data is used. Specifically, while BLEU-4 exhibits a 51.8% improvement over the CLIP model in a zero-shot setting, this improvement diminishes to 25.0% when the full dataset is utilized. This suggests that the benefits of prompts decrease as models learn from sufficient examples, yielding diminishing returns.

4.6. Qualitative analysis

We further provide some examples to show the effectiveness of our model. Fig. 3 gives the reports in different languages and different modalities generated by ZeroMRG. We can find that our model is able to comprehend and incorporate domain-specific medical terminology along with the relevant context, ensuring the relevance and clinical accuracy of the generated reports. For instance, in the first Chinese medical report, it states, "双肺下叶胸膜下见多发斑片状磨玻璃影, 病灶内小叶间隔增厚" while in the second Chinese medical report, it notes, "见若干细小淋巴结, 双侧腋窝淋巴结稍大, 短径约8 mm", etc. In the first English medical report, it states, *"Patchy solid shadow was seen in the upper lobe of the left lung"* while in the second English medical report, it notes, *"Two nodules were seen in the lingual segment of the upper lobe of the left lung"*. In the third English medical report, it states, *"Calcified granuloma identified in the right suprahilar region"* while in the last English medical report, it notes, *"Heart size mildly to moderately enlarged"*. In detail, our method correctly captures the abnormalities like "玻璃影", "淋巴结", *"shadow"*, *"nodules"*, *"calcified"*, *"granuloma"*, which is crucial for medical report generation.

5. Conclusion

In this paper, we introduce a novel approach, the ZeroMRG, which is the attempt to achieve zero-shot multimodal and multilingual medical report generation within a unified framework. By leveraging autoencoding and prompt learning tailored specifically for report generation, even in the absence of paired images and reports, ZeroMRG demonstrates remarkable adaptability and versatility. ZeroMRG represents a significant step forward in advancing the state-of-the-art in medical report generation and holds promise for broader applications in healthcare informatics.

However, our study has limitations. Firstly, the model's performance relies on the quality of pre-trained multilingual embeddings, which may not cover low-resource languages equally. Secondly, while ZeroMRG generates coherent reports, fine-grained anatomical accuracy in complex cases (e.g., rare diseases) requires further validation by clinicians. Thirdly, deploying ZeroMRG in real-world settings faces challenges such as integrating with hospital systems and ensuring compliance with regional data privacy regulations.

For practical implementation, we envision ZeroMRG being used in two key scenarios: firstly, as a diagnostic aid in resource-limited regions where radiologists are scarce, providing preliminary reports in local languages; secondly, supporting cross-border telemedicine by producing standardized reports in multiple languages for international specialist consultations. In the future, we will continue to explore zero-shot models for an even broader range of languages and modalities. To bridge the performance gap for low-resource languages, we will develop adaptive embedding architectures and overcome the limitations of Multilingual-CLIP. Additionally, we will delve into the application of zero-shot learning in other domains.

CRediT authorship contribution statement

Yuena Jiang: Investigation, Writing – original draft, Formal analysis, Validation, Methodology. **Yanxun Chang:** Supervision, Funding acquisition, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

References

Alfarghaly, O., Khaled, R., Elkorany, A., et al., 2021. Automated radiology report generation using conditioned transformers. Inform. Med. Unlocked 24, 100557.
Bhardwaj, P., Bhat, S., Maier, A., 2025. Enhancing zero-shot learning in chest X-ray diagnostics. In: Bildverarbeitung Für Die Medizin 2025: Proceedings, German Conference on Medical Image Computing, Regensburg March 09-11, 2025. Springer-Verlag, p. 270.
Bottou, L., Curtis, F.E., Nocedal, J., 2018. Optimization methods for large-scale machine learning. SIAM Rev. 60 (2), 223–311.
Brown, T.B., Mann, B., Ryder, N., et al., 2020. Language models are few-shot learners. arXiv preprint arXiv:2005.14165.
Carlsson, F., Eisen, P., Rekathati, F., et al., 2022. Cross-lingual and multilingual clip. In: Proceedings of the Thirteenth Language Resources and Evaluation Conference. pp. 6848–6854.
Chen, X., Fang, H., Lin, T.-Y., et al., 2015. Microsoft coco captions: Data collection and evaluation server. arXiv preprint arXiv:1504.00325.
Chen, J., Guo, H., Yi, K., et al., 2022a. Visualgpt: Data-efficient adaptation of pretrained language models for image captioning. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition. pp. 18030–18040.
Chen, Z., Shen, Y., Song, Y., et al., 2022b. Cross-modal memory networks for radiology report generation. arXiv preprint arXiv:2204.13258.
Chen, Z., Song, Y., Chang, T.-H., et al., 2020. Generating radiology reports via memory-driven transformer. arXiv preprint arXiv:2010.16056.
Cornia, M., Stefanini, M., Baraldi, L., Cucchiara, R., 2020b. Meshed-memory transformer for image captioning. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition. pp. 10578–10587.
Cornia, M., Stefanini, M., Baraldi, L., et al., 2020a. Meshed-memory transformer for image captioning. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition. pp. 10578–10587.
Demner-Fushman, D., Kohli, M.D., Rosenman, M.B., Shooshan, S.E., Rodriguez, L., Antani, S., Thoma, G.R., McDonald, C.J., 2016. Preparing a collection of radiology examinations for distribution and retrieval. J. Am. Med. Inform. Assoc. 23 (2), 304–310.
Denkowski, M., Lavie, A., 2014. Meteor universal: Language specific translation evaluation for any target language. In: Proceedings of the Ninth Workshop on Statistical Machine Translation. pp. 376–380.
Devlin, J., Chang, M.-W., Lee, K., et al., 2018. Bert: Pre-training of deep bidirectional transformers for language understanding. arXiv preprint arXiv:1810.04805.
Goergen, S.K., Pool, F.J., Turner, T.J., et al., 2013. Evidence-based guideline for the written radiology report: Methods, recommendations and implementation challenges. J. Med. Imaging Radiat. Oncol. 57 (1), 1–7.
Gu, Y., Han, X., Liu, Z., et al., 2021. PPT: Pre-trained prompt tuning for few-shot learning. arXiv preprint arXiv:2109.04332.
Gu, J., Joty, S., Cai, J., et al., 2018. Unpaired image captioning by language pivoting. In: Proceedings of the European Conference on Computer Vision. pp. 503–519.
Hirsch, E., Dawidowicz, G., Tal, A., 2024. MedRAT: Unpaired medical report generation via auxiliary tasks. arXiv preprint arXiv:2407.03919.
Hu, D., Liu, B., Zhu, X., Lu, X., Wu, N., 2024. Zero-shot information extraction from radiological reports using ChatGPT. Int. J. Med. Inform. 183, 105321.
Huang, W., Li, C., Zhou, H.-Y., Yang, H., Liu, J., Liang, Y., Zheng, H., Zhang, S., Wang, S., 2024. Enhancing representation in radiography-reports foundation model: A granular alignment algorithm using masked contrastive learning. Nat. Commun. 15 (1), 7620.
Javaheripi, M., Bubeck, S., Abdin, M., et al., 2023. Phi-2: The surprising power of small language models. URL: https://www.microsoft.com/en-us/research/blog/phi-2-the-surprising-power-of-small-language-models.
Jing, B., Xie, P., Xing, E., 2017. On the automatic generation of medical imaging reports. arXiv preprint arXiv:1711.08195.
Kumar, A., Irsoy, O., Ondruska, P., et al., 2016. Ask me anything: Dynamic memory networks for natural language processing. In: International Conference on Machine Learning. pp. 1378–1387.
Lee, J., Yoon, W., Kim, S., et al., 2020. BioBERT: a pre-trained biomedical language representation model for biomedical text mining. Bioinform. 36 (4), 1234–1240.
Li, Y., Liang, X., Hu, Z., et al., 2018. Hybrid retrieval-generation reinforced agent for medical image report generation. Adv. Neural Inf. Process. Syst. 31, 1537–1547.
Li, M., Liu, R., Wang, F., et al., 2023b. Auxiliary signal-guided knowledge encoder-decoder for medical report generation. World Wide Web 26 (1), 253–270.
Li, C., Wong, C., Zhang, S., et al., 2023a. Llava-med: Training a large language-and-vision assistant for biomedicine in one day. arXiv preprint arXiv:2306.00890.
Li, W., Zhu, L., Wen, L., et al., 2023c. Decap: Decoding clip latents for zero-shot captioning via text-only training. arXiv preprint arXiv:2303.03032.
Lin, C.-Y., 2004. Rouge: A package for automatic evaluation of summaries. In: Text Summarization Branches Out. pp. 74–81.
Liu, F., Ge, S., Wu, X., 2022a. Competence-based multimodal curriculum learning for medical report generation. arXiv preprint arXiv:2206.14579.
Liu, G., Hsu, T.-M.H., McDermott, M., et al., 2019. Clinically accurate chest x-ray report generation. In: Machine Learning for Healthcare Conference. pp. 249–269.
Liu, R., Li, M., Zhao, S., Chen, L., Chang, X., Yao, L., 2024b. In-context learning for zero-shot medical report generation. In: Proceedings of the 32nd ACM International Conference on Multimedia. pp. 8721–8730.
Liu, G., Liao, Y., Wang, F., et al., 2021d. Medical-vlbert: Medical visual language bert for covid-19 ct report generation with alternate learning. IEEE Trans. Neural Netw. Learn. Syst. 32 (9), 3786–3797.
Liu, C., Tian, Y., Chen, W., et al., 2024a. Bootstrapping large language models for radiology report generation. In: Proceedings of the AAAI Conference on Artificial Intelligence. Vol. 38, pp. 18635–18643.
Liu, F., Wu, X., Ge, S., et al., 2021a. Exploring and distilling posterior and prior knowledge for radiology report generation. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition. pp. 13753–13762.
Liu, F., Wu, X., You, C., et al., 2022b. Aligning source visual and target language domains for unpaired video captioning. IEEE Trans. Pattern Anal. Mach. Intell. 44 (12), 9255–9268.
Liu, F., Yin, C., Wu, X., et al., 2021b. Contrastive attention for automatic chest x-ray report generation. arXiv preprint arXiv:2106.06965.
Liu, F., You, C., Wu, X., et al., 2021c. Auto-encoding knowledge graph for unsupervised medical report generation. Adv. Neural Inf. Process. Syst. 34, 16266–16279.
Liu, F., Zhu, T., Wu, X., et al., 2023. A medical multimodal large language model for future pandemics. NPJ Digit. Med. 6 (1), 226.
Lu, J., Xiong, C., Parikh, D., et al., 2017. Knowing when to look: Adaptive attention via a visual sentinel for image captioning. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition. pp. 375–383.
Monshi, M.M.A., Poon, J., Chung, V., 2020. Deep learning in generating radiology reports: A survey. Artif. Intell. Med. 106, 101878.
Nukrai, D., Mokady, R., Globerson, A., 2022. Text-only training for image captioning using noise-injected clip. arXiv preprint arXiv:2211.00575.
Pan, Y., Yao, T., Li, Y., et al., 2020. X-linear attention networks for image captioning. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition. pp. 10971–10980.
Papineni, K., Roukos, S., Ward, T., et al., 2002. Bleu: a method for automatic evaluation of machine translation. In: Proceedings of the 40th Annual Meeting of the Association for Computational Linguistics. pp. 311–318.
Pool, F., Goergen, S., 2010. Quality of the written radiology report: a review of the literature. J. Am. Coll. Radiol. 7 (8), 634–643.
Qin, H., Song, Y., 2022. Reinforced cross-modal alignment for radiology report generation. In: Findings of the Association for Computational Linguistics. pp. 448–458.
Radford, A., Kim, J.W., Hallacy, C., et al., 2021. Learning transferable visual models from natural language supervision. In: International Conference on Machine Learning. pp. 8748–8763.
Radford, A., Narasimhan, K., Salimans, T., et al., 2018. Improving language understanding by generative pre-training. URL: https://s3-us-west-2.ama-zonaws.com/openaiassets/researchcovers/languageunsupervised/languageunderstandingpaper.pdf.

Rennie, S.J., Marcheret, E., Mroueh, Y., Ross, J., Goel, V., 2017. Self-critical sequence training for image captioning. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition. pp. 7008–7024.

Selivanov, A., Rogov, O.Y., Chesakov, D., et al., 2023. Medical image captioning via generative pretrained transformers. Sci. Rep. 13 (1), 4171.

Shin, H.-C., Roberts, K., Lu, L., et al., 2016. Learning to read chest x-rays: Recurrent neural cascade model for automated image annotation. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition. pp. 2497–2506.

Tanida, T., Müller, P., Kaissis, G., et al., 2023. Interactive and explainable region-guided radiology report generation. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition. pp. 7433–7442.

Tewel, Y., Shalev, Y., Schwartz, I., et al., 2022. Zerocap: Zero-shot image-to-text generation for visual-semantic arithmetic. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition. pp. 17918–17928.

Thawkar, O., Shaker, A., Mullappilly, S.S., et al., 2023. Xraygpt: Chest radiographs summarization using medical vision-language models. arXiv preprint arXiv:2306.07971.

Touvron, H., Lavril, T., Izacard, G., et al., 2023. Llama: Open and efficient foundation language models. arXiv preprint arXiv:2302.13971.

Tu, T., Azizi, S., Driess, D., et al., 2023. Towards generalist biomedical ai. arXiv preprint arXiv:2307.14334.

Vaswani, A., Shazeer, N., Parmar, N., et al., 2017. Attention is all you need. Adv. Neural Inf. Process. Syst. 30, 5998–6008.

Vedantam, R., Lawrence Zitnick, C., Parikh, D., 2015. Cider: Consensus-based image description evaluation. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition. pp. 4566–4575.

Vinyals, O., Blundell, C., Lillicrap, T., et al., 2016. Matching networks for one shot learning. Adv. Neural Inf. Process. Syst. 29, 3630–3638.

Vinyals, O., Toshev, A., Bengio, S., et al., 2015. Show and tell: A neural image caption generator. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition. pp. 3156–3164.

Waisberg, E., Ong, J., Masalkhi, M., et al., 2023. GPT-4: a new era of artificial intelligence in medicine. Ir. J. Med. Sci. 192 (6), 3197–3200.

Wang, W., Chen, Z., Hu, H., 2019. Hierarchical attention network for image captioning. In: Proceedings of the AAAI Conference on Artificial Intelligence. Vol. 33, pp. 8957–8964.

Wang, X., Peng, Y., Lu, L., et al., 2018. Tienet: Text-image embedding network for common thorax disease classification and reporting in chest x-rays. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition. pp. 9049–9058.

Xu, K., Ba, J., Kiros, R., et al., 2015. Show, attend and tell: Neural image caption generation with visual attention. In: International Conference on Machine Learning. pp. 2048–2057.

Yang, B., Liu, F., Zou, Y., et al., 2024. Zeronlg: Aligning and autoencoding domains for zero-shot multimodal and multilingual natural language generation. IEEE Trans. Pattern Anal. Mach. Intell. 46, 5712–5724.

Yang, X., Tang, K., Zhang, H., et al., 2019. Auto-encoding scene graphs for image captioning. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition. pp. 10685–10694.

Yuan, J., Liao, H., Luo, R., et al., 2019. Automatic radiology report generation based on multi-view image fusion and medical concept enrichment. In: Medical Image Computing and Computer Assisted Intervention. pp. 721–729.

Zeng, A., Attarian, M., Ichter, B., et al., 2022. Socratic models: Composing zero-shot multimodal reasoning with language. arXiv preprint arXiv:2204.00598.

Zhang, Y., Wang, X., Xu, Z., et al., 2020. When radiology report generation meets knowledge graph. In: Proceedings of the AAAI Conference on Artificial Intelligence. Vol. 34, pp. 12910–12917.

Zhang, Z., Xie, Y., Xing, F., et al., 2017. Mdnet: A semantically and visually interpretable medical image diagnosis network. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition. pp. 6428–6436.

Zhou, K., Yang, J., Loy, C.C., et al., 2022. Learning to prompt for vision-language models. Int. J. Comput. Vis. 130 (9), 2337–2348.

Zhu, D., Chen, J., Shen, X., et al., 2023. Minigpt-4: Enhancing vision-language understanding with advanced large language models. arXiv preprint arXiv:2304.10592.